Chapter 3 – VirtualBox Networking & Lab Design

Before installing Wazuh, all virtual machines (VMs) must be able to communicate with each other.

If the network is configured incorrectly:

- The Wazuh Agent cannot connect to the Manager.
- Kali Linux cannot attack the Windows machine.
- Ubuntu Server cannot receive logs.
- The Dashboard may not be accessible.

Therefore, understanding VirtualBox networking is essential.

3.1 What is VirtualBox?

VirtualBox is a Type-2 Hypervisor developed by Oracle.

It allows you to run multiple virtual computers (Virtual Machines) on one physical computer.

Example:

```
Your Physical PC

│

├── Ubuntu Server VM
├── Windows 11 VM
├── Kali Linux VM
└── Windows Server VM (Optional)
```

Each VM behaves like a separate computer with its own:

- CPU

- RAM
- Storage
- Network Adapter
- Operating System

3.2 What is a Virtual Network Adapter?

A Virtual Network Adapter works like a physical Network Interface Card (NIC).

It allows a VM to:

- Communicate with other VMs
- Access the Internet
- Connect to the Host Computer

Every VM can have multiple network adapters.

3.3 VirtualBox Network Modes

VirtualBox provides several networking modes.

The most important ones are:

- NAT
- Bridged Adapter
- Host-Only Adapter
- Internal Network

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1. NAT (Network Address Translation)

Definition

NAT allows the VM to access the Internet through the Host Computer.

The outside world cannot directly connect to the VM.

Diagram

Internet→Host Computer→VirtualBox NAT→Ubuntu VM

Advantages

- Easy to configure
- Internet access works immediately
- Safe for beginners

Disadvantages

- Other devices cannot directly reach the VM.
- Not ideal for enterprise labs.

Best Use

- Downloading software
- Updating Ubuntu
- Installing packages

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2. Bridged Adapter

Definition

A Bridged Adapter connects the VM directly to the physical network.

The VM receives its own IP address from the router.

Diagram

Router

├── Host PC

├── Ubuntu Server

├── Windows 11

└── Kali Linux

Every machine appears as a separate computer on the network.

Advantages

- Full network communication
- Easy remote access
- Works like a real enterprise network

Disadvantages

- Requires access to the physical network.
- Less isolated than Host-Only mode.

Best Use

- Enterprise labs
- SSH access
- Realistic networking

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3. Host-Only Adapter

Definition

Host-Only networking creates a private network between the Host Computer and the Virtual Machines.

The VMs cannot access the Internet.

Diagram

Host Computer

│

VirtualBox Host Network

├── Ubuntu

├── Windows

└── Kali

Advantages

- Completely isolated
- Safe for malware testing
- Excellent for cybersecurity labs

Disadvantages

- No Internet connection

Best Use

- Malware Analysis
- Attack Simulations
- Security Testing

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4. Internal Network

Definition

Internal Network creates a network only between Virtual Machines.

Even the Host Computer cannot access them.

Diagram

Ubuntu

|

Internal Network

├── Windows

└── Kali

Advantages

- Maximum isolation
- Realistic enterprise simulation

Disadvantages

- Difficult for beginners
- No Host access

Best Use

- Advanced Labs
- Malware Research

3.4 Which Network Should We Use?

For our Wazuh Home Lab, we will use **two network adapters** on each VM.

Adapter 1

NAT

Purpose:

- Internet Access
- Ubuntu Updates
- Package Installation
- Downloading Wazuh

Adapter 2

Host-Only Adapter

Purpose:

- Communication between Ubuntu Server, Windows, and Kali
- Safe attack simulation
- Wazuh Agent communication
- No exposure to your home network

3.5 Home Lab Network Design

```
          Internet
             |
         Home Router
             |
        Host Computer
             |
       VirtualBox (NAT)
             |
        Ubuntu Server
      (Internet Access)
```

Host-Only Network

```
192.168.56.0/24

    │

├── Ubuntu Server
│     192.168.56.10
│
├── Windows 11
│     192.168.56.20
│
└── Kali Linux
      192.168.56.30
```

The **Host-Only network** is where all cybersecurity activities occur.

3.6 Why Two Network Adapters?

Using two adapters separates responsibilities.

Adapter 1 (NAT)

Used for:

- Installing software
- Updating Linux
- Downloading packages

Adapter 2 (Host-Only)

Used for:

- Wazuh communication

- Agent connections
- Nmap scans
- Attack simulations
- Log collection

This design is commonly used in home labs because it is both practical and secure.

3.7 IP Address Plan

We will assign static IP addresses.

Machine	IP Address
Ubuntu Server	192.168.56.10
Windows 11	192.168.56.20
Kali Linux	192.168.56.30

Using static IP addresses ensures the Wazuh Agent always knows where to find the Wazuh Manager.

3.8 Data Flow

Kali Linux→Attack→Windows 11→Windows Logs→Wazuh Agent→Ubuntu Server→Wazuh Manager→Indexer→Dashboard→SOC Analyst

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3.9 Why We Do Not Install Wazuh on Kali

Kali Linux is designed for offensive security.

It should remain the attack machine.

Ubuntu Server is optimized to host services such as:

- Wazuh Manager
- Wazuh Dashboard

- Wazuh Indexer

Separating these roles creates a more realistic enterprise lab.

Chapter 3 Summary

VirtualBox allows multiple operating systems to run on one physical computer.

The four main networking modes are:

- NAT
- Bridged Adapter
- Host-Only Adapter
- Internal Network

For this Wazuh Home Lab, we will use:

- **Adapter 1:** NAT (Internet access)
- **Adapter 2:** Host-Only (private lab network)

This setup allows us to safely install Wazuh, monitor Windows, simulate attacks from Kali, and investigate alerts without affecting the home network.

One-Line Summary

Use NAT for Internet access and Host-Only for secure communication between the Ubuntu Server, Windows 11, and Kali Linux virtual machines in the Wazuh home lab.